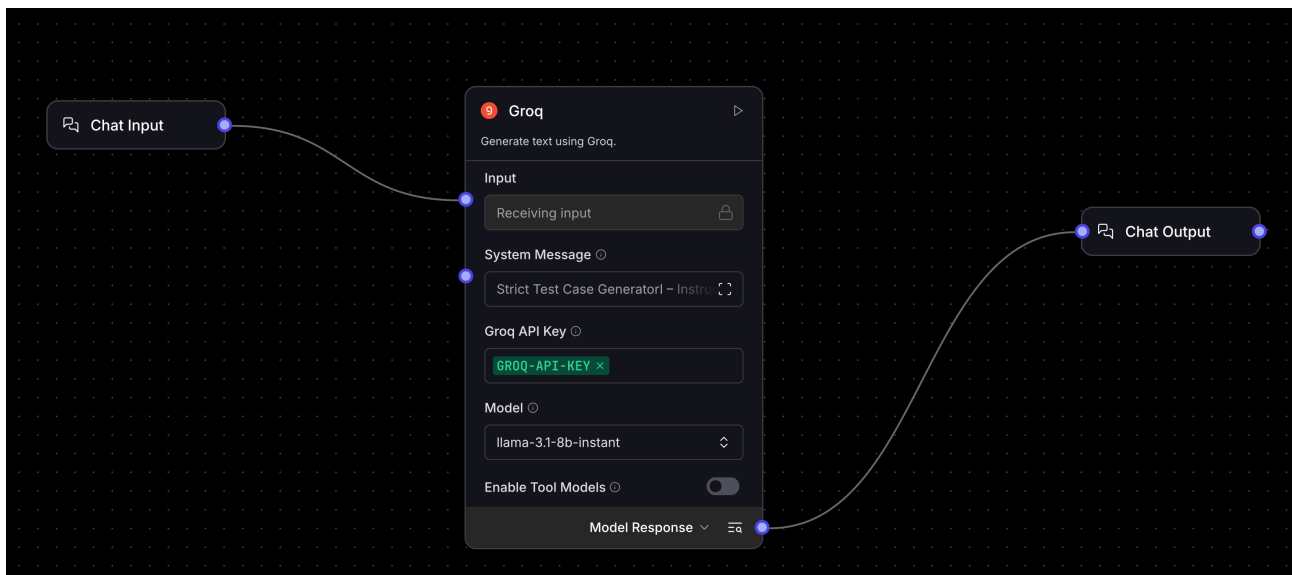


IMPLEMENTATION 1: TEST CASE GENERATOR THROUGH LANGFLOW CDN

STEP 1

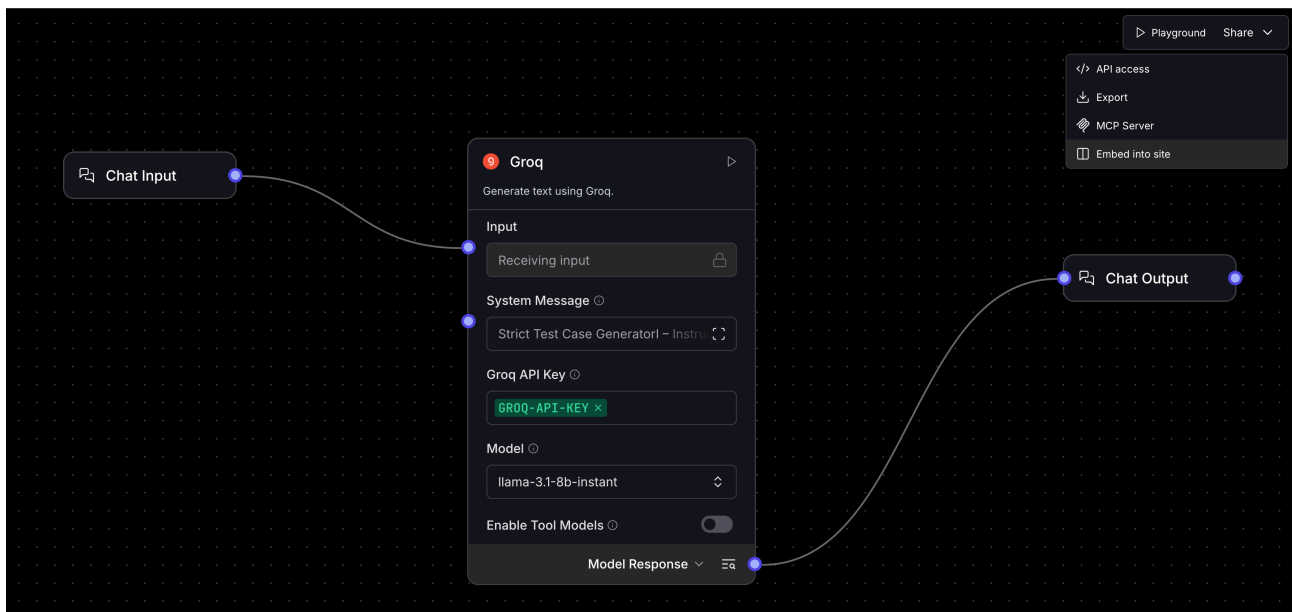
Create a Langflow workflow pipeline that generates a test case for a given input. The pipeline contains three components

- Chat Input that takes in the user's instruction
- A GROQ component that takes in the user input as an INPUT message, and the instructions are provided via an ICEPOT prompt in a system message
- Finally, a Chat Output component that displays the generated test cases as a response in the chat UI



STEP 2

For the created workflow, click on the **Share** option at the top right and click on **Embed into site**



STEP 3

Copy the generated HTML block into a text file and save the file as a **.html** document

Content from **Embed into site**

```
Embed into site

1 <script
2   src="https://cdn.jsdelivr.net/gh/langflow-ai/langflow-embedded-chat@v1.0.8/dist/build/static/js/bundle.min.js">
3 </script>
4 <langflow-chat
5   window_title="01_P2W1D1_ClassroomAssignment-TestCaseGenerator"
6   flow_id="c2c7be4c-8489-4b39-a2ca-8ec765172a1d"
7   host_url="http://localhost:7860">
8 </langflow-chat>
```

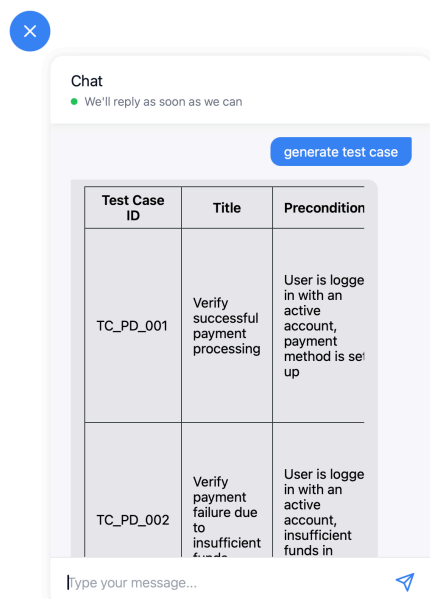
Content from **Embed into site** saved in an **.html** file

```
chat_Implementation1.html x  Untitled.pages .../week-1/...  Untitled.pages .../week-2/day-1 U  Untitled.pages .../day-2

phase-2 > week-2 > day-1 > chat_Implementation1.html > ...
1 <html lang="en">
2 <head>
3 <script src="https://cdn.jsdelivr.net/gh/langflow-ai/langflow-embedded-chat@v1.0.7/dist/build/static/js/bundle.min.js"></script>
4 </head>
5 <body>
6 <langflow-chat
7   host_url="http://localhost:7860"
8   flow_id="c2c7be4c-8489-4b39-a2ca-8ec765172a1d"
9   api_key="sk-ppT_yvwWqyRRluQ1qBxMuvX5BKaiWiJL7JhAGWlfMCA"
10 ></langflow-chat>
11 </body>
12 </html>
13 |
```

STEP 4

Loading **.html** document on a browser loads the Chatbot UI. The UI utilizes Langflow CDN **https://cdn.jsdelivr.net/gh/langflow-ai/langflow-embedded-chat@v1.0.7/dist/build/static/js/bundle.min.js** and uses the Flow ID, Langflow deployed URL, and Langflow API key to interact with the UI with the workflow

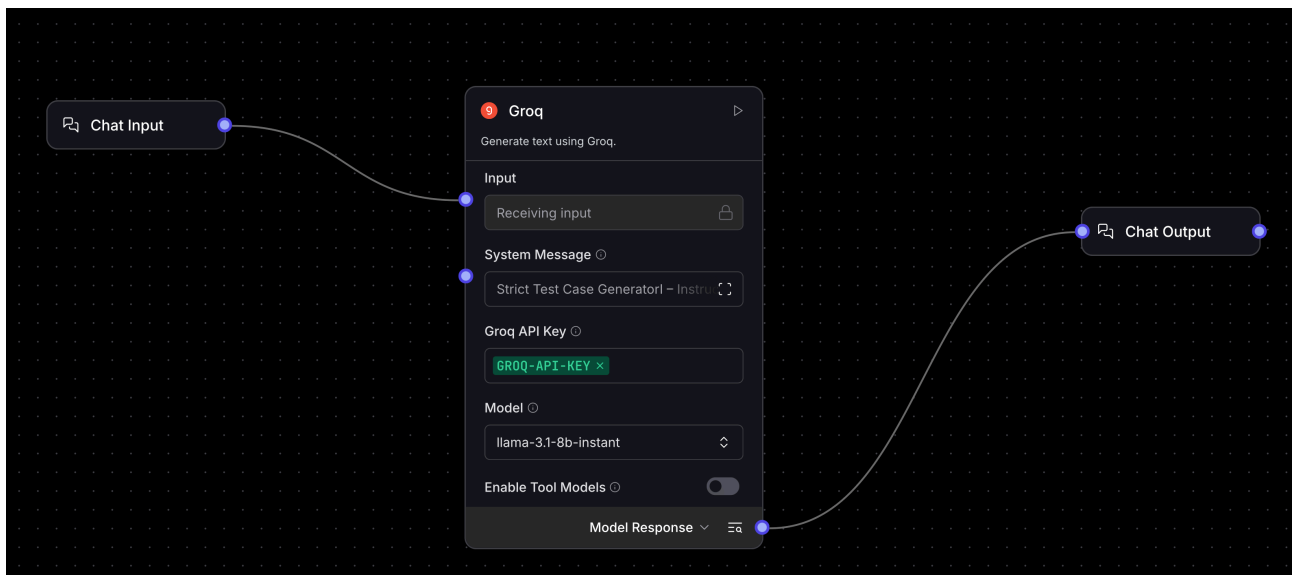


IMPLEMENTATION 2: TEST CASE GENERATOR CUSTOMIZATION THROUGH LANGFLOW CDN

STEP 1

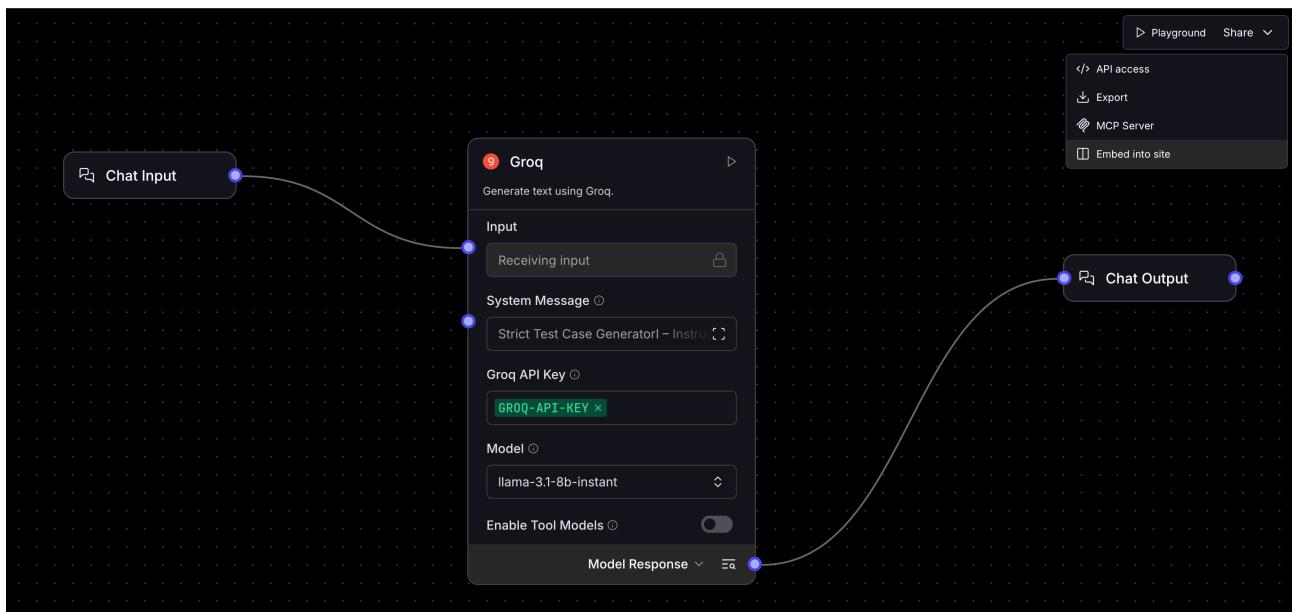
Create a Langflow workflow pipeline that generates a test case for a given input. The pipeline contains three components

- Chat Input that takes in the user's instruction
- A GROQ component that takes in the user input as an INPUT message, and the instructions are provided via an ICEPOT prompt in a system message
- Finally, a Chat Output component that displays the generated test cases as a response in the chat UI



STEP 2

For the created workflow, click on the **Share** option at the top right and click on **Embed into site**



STEP 3

Copy the generated HTML block into a text file and save the file as a **.html** document

Content from **Embed into site**

```
Embed into site

1 <script
2   src="https://cdn.jsdelivr.net/gh/langflow-ai/langflow-embedded-chat@v1.0.8/dist/build/static/js/bundle.min.js">
3 </script>
4 <langflow-chat
5   window_title="01_P2W1D1_ClassroomAssignment-TestCaseGenerator"
6   flow_id="c2c7be4c-8489-4b39-a2ca-8ec765172a1d"
7   host_url="http://localhost:7860">
8 </langflow-chat>
```

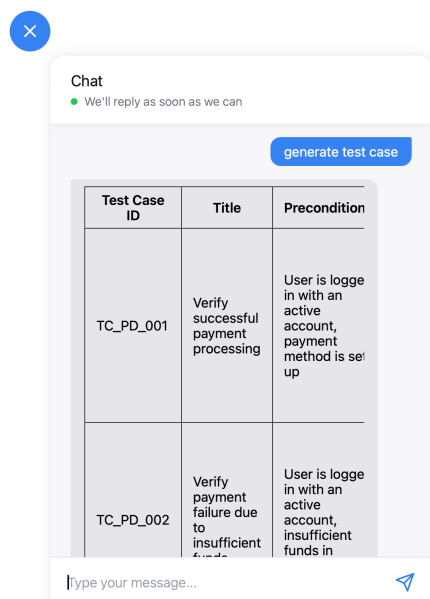
Content from **Embed into site** saved in an **.html** file

```
chat_Implementation1.html x  Untitled.pages .../week-1/...  Untitled.pages .../week-2/day-1 U  Untitled.pages .../day-2

phase-2 > week-2 > day-1 > chat_Implementation1.html > ...
1 <html lang="en">
2 <head>
3 <script src="https://cdn.jsdelivr.net/gh/langflow-ai/langflow-embedded-chat@v1.0.7/dist/build/static/js/bundle.min.js"></script>
4 </head>
5 <body>
6 <langflow-chat
7   host_url="http://localhost:7860"
8   flow_id="c2c7be4c-8489-4b39-a2ca-8ec765172a1d"
9   api_key="sk-ppT_yvwWqyRRluQ1qBxMuvX5BKaiWiJL7JhAGWlfMCA"
10 ></langflow-chat>
11 </body>
12 </html>
13 |
```

STEP 4

Loading **.html** document on a browser loads the Chatbot UI. The UI utilizes Langflow CDN **https://cdn.jsdelivr.net/gh/langflow-ai/langflow-embedded-chat@v1.0.7/dist/build/static/js/bundle.min.js** and uses the Flow ID, Langflow deployed URL, and Langflow API key to interact with the UI with the workflow



STEP 5

UI customizations such as background color, text color, chatbot position, etc., can be configured by modifying the HTML block.

In the HTML block below, chat position, background color, trigger style, etc., have been modified

```
phase-2 > week-2 > day-1 > chat_Implementation2_CustomizationOfChat.html > ...
1 <html lang="en">
2 <head>
3 <script src="https://cdn.jsdelivr.net/gh/langflow-ai/langflow-embedded-chat@v1.0.7/dist/build/static/js/bundle.min.js"></script>
4 </head>
5 <body>
6 <langflow-chat
7   host_url="http://localhost:7860"
8   flow_id="c2c7be4c-8489-4b39-a2ca-8ec765172a1d"
9   api_key="sk-ppT_yvwVqYRRLu01qBxMuuxS8KaIWjL7jhAGWLFMCA"
10  bot_message_style='{ "color": "#ffffff", "backgroundColor": "#3B82F6" }'
11  chat_position="bottom-right"
12  chat_trigger_style='{ "backgroundColor": "#3B82F6", "color": "#fff" }'
13  online=true
14  chat_window_style='{ "borderRadius": "8px", "boxShadow": "0 2px 8px rgba(0,0,0,0.1)" }'
15 ></langflow-chat>
16 </body>
17 </html>
18
```



Chat

● We'll reply as soon as we can

Generate Test Case

Test Case ID	Title	Precondition
TC_PD_001	Verify successful payment for patient services	User is logged in with an active account, patient services are available for payment
		User is logged in with an

Type your message...

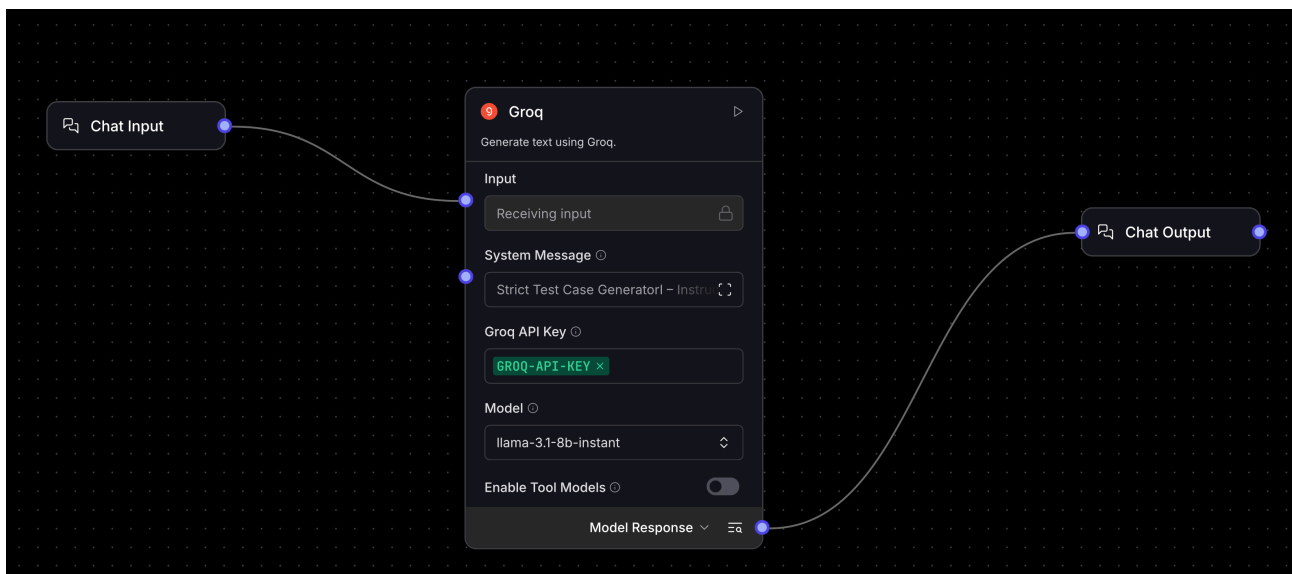


IMPLEMENTATION 3: TEST CASE GENERATOR BUILT ON HTML/CSS UI THROUGH LANGFLOW CDN AS BACKEND

STEP 1

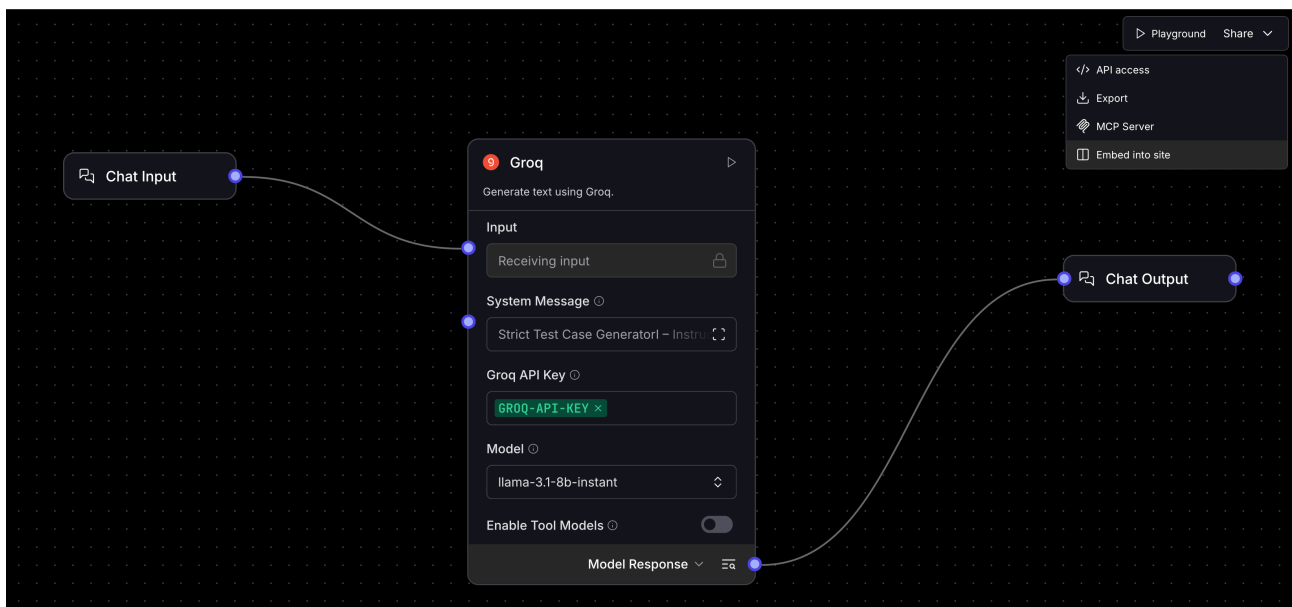
Create a Langflow workflow pipeline that generates a test case for a given input. The pipeline contains three components

- Chat Input that takes in the user's instruction
- A GROQ component that takes in the user input as an INPUT message, and the instructions are provided via an ICEPOT prompt in a system message
- Finally, a Chat Output component that displays the generated test cases as a response in the chat UI



STEP 2

For the created workflow, click on the **Share** option at the top right and click on **Embed into site**



STEP 3

Copy the generated HTML block into a text file and save the file as a **.html** document

Content from **Embed into site**

```
Embed into site

1 <script
2   src="https://cdn.jsdelivr.net/gh/langflow-ai/langflow-embedded-chat@v1.0.8/dist/build/static/js/bundle.min.js">
3 </script>
4 <langflow-chat
5   window_title="01_P2W1D1_ClassroomAssignment-TestCaseGenerator"
6   flow_id="c2c7be4c-8489-4b39-a2ca-8ec765172a1d"
7   host_url="http://localhost:7860">
8 </langflow-chat>
```

STEP 4

- Within the local system, create a project folder and open it in the VS Code workspace.
- Connect with an AI agent such as Copilot / Claude and provide a detailed instruction prompt specifying the implementation details
- Proceed Step By Step, validating each implemented section, and proceed to the next set of implementation only when the existing implementation is completed as expected

During the exercise, the following prompts were utilized

Version 1

Context

- Act as a senior developer and create a web application that generates test cases.

Instructions

- MANDATORY - The application needs to have a Title named Test Case Generator
- MANDATORY - The application should have an option of generating test cases for frequently used workflows for the Netbanking, Financial Services, and Insurance Domain
- MANDATORY - The Application should contain a chat agent where the user can interact with the agent to generate test cases using a chat prompt
- STRICTLY - USE the Langflow chat agent pipeline that has been developed.
- STRICTLY - USE LangFlow CDN to interact with the LangFlow agent. The CDN details are provided below
 - ENDPOINT: `http://localhost:7860/api/v1/run`
 - LangFlow API KEY: `sk-hhPrnyU51OZWtd0bTk3yulZNNP4drSxGjfd-ihU-rsE`
 - Flow ID: `e6248328-6b44-4366-8738-f846364f19fe`
 - METHOD: POST
 - Content of Langflow CDN that needs to be embedded into UI

```
<script
  src="https://cdn.jsdelivr.net/gh/langflow-ai/langflow-embedded-chat@v1.0.8/dist/build/static/js/bundle.min.js">
</script>
<langflow-chat
  window_title="01_P2W1D1_ClassroomAssignment-TestCaseGenerator"
  flow_id="c2c7be4c-8489-4b39-a2ca-8ec765172a1d"
  host_url="http://localhost:7860">
</langflow-chat>
```
- MANDATORY - Keep all CDN details, such as CDN URL, flow ID, and endpoint details, in a config file/env file

- 8) MANDATORY - Provide the user with the option to download the generated test cases in Excel format
- 9) MANDATORY - Keep a toggle for changing between dark and light mode

Version 2

The following instructions were provided as customization instructions to make the UI interactive and appealing

Can you make some modifications to the UI.

- a) The page title Test Case Generator needs to be changed. Think of an innovative name using your creative skills. Set the title not on top left and in the center. Also make the title banner blue theme with the text in White.
- b) Move the Quick Generate Pane to the Extrem left. Also do not show any of the dummy test cases by default. Keep it hidden and once a user click on the Domain type, the dummy test cases will load within a block for that specific domain
- c) Expand the Chat section having the below DOM structure to occupy the rest of the screen

```
<section class="chat-panel">
  <div class="chat-window" id="chat-window">
    <div class="chat-message agent">
      <div class="bubble">
        🙋 Hi! I'm your Test Case Generator agent. Pick a workflow on the left, or type a
        request below to generate test cases for Netbanking, Financial Services, or
        Insurance domains.
      </div>
    </div>
  </div>
</section>
```

```
    <div class="chat-input-row">
      <textarea id="chat-input" placeholder="Ask me to generate test cases... e.g. 'Generate test
cases for fund transfer'" rows="2"></textarea>
      <button id="send-btn" class="primary-btn">Send</button>
    </div>
```

```
    <div class="action-row">
      <button id="download-btn" class="secondary-btn" disabled="">
        ↓ Download Test Cases (Excel)
      </button>
      <span id="status-text" class="status-text"></span>
    </div>
  </section>
```

- d) For the Download Excel part, instead of having full text specifying Download Test Cases (Excel), add a excel icon image and a export icon image to visually make an user undersatn that excel export can be performed using that section. I do not want full text mentioned

FINAL OUTOUT

Once the implementation is completed, this is how the UI looks. It contains a left pane that provides the user with the option to generate test cases for frequently executed workflows in the Netbanking, Financial Services, and Insurance domains

CaseForge AI

Quick Generate

Click a domain to reveal its frequently used workflows.

Netbanking

Financial Services

Insurance

New policy purchase

Claim filing and tracking

Premium payment

Policy renewal

Nominee addition / update

Policy document download

👋 Hi! I'm your Test Case Generator agent. Pick a workflow on the left, or type a request below to generate test cases for Netbanking, Financial Services, or Insurance domains.

Ask me to generate test cases... e.g. 'Generate test cases for fund transfer'

Send

📎

On clicking any of the frequently used test cases, it interacts with the chat bot test case generator and displays the generated test case in the chat UI

CaseForge AI

Quick Generate

Click a domain to reveal its frequently used workflows.

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Financial Services

Insurance

New policy purchase

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👋 Hi! I'm your Test Case Generator agent. Pick a workflow on the left, or type a request below to generate test cases for Netbanking, Financial Services, or Insurance domains.

Ask me to generate test cases... e.g. 'Generate test cases for fund transfer'

Send

📎

The user can provide custom instructions in the chat text box to generate a customized test case for the Netbanking, Financial Services, and Insurance domains. The user can also download the generated test case in Excel format

CaseForge AI

Quick Generate

Click a domain to reveal its frequently used workflows.

Netbanking

Financial Services

Insurance

New policy purchase

Claim filing and tracking

Premium payment

Policy renewal

Nominee addition / update

Policy document download

in linked payment method | 1. Navigate to "Buy New Policy".->2. Select "Health Insurance".->3. Choose a plan and enter required personal details.->4. Enter a premium amount within the allowed range (e.g., ₹5,000-₹50,000).->5. Confirm policy terms and click "Proceed".->6. Receive OTP on registered mobile, enter OTP.->7. Authorize payment.->8. Click "Confirm Purchase". | Policy is created, payment is captured, and a confirmation screen with policy number and receipt is displayed. |

ITC_IP_002 | Purchase fails when premium exceeds maximum allowed limit | User is logged in; user has a verified mobile number | 1. Navigate to "Buy New Policy".->2. Select "Motor Insurance".->3. Enter vehicle details.->4. Input a premium amount above the maximum allowed (e.g., ₹2,00,000 when limit is ₹1,50,000).->5. Click "Proceed". | System displays a validation error: "Premium exceeds maximum allowed limit" and prevents further progression. |

ITC_IP_003 | Validation error for missing mandatory fields | User is logged in | 1. Navigate to "Buy New Policy".->2. Select "Life Insurance".->3. Leave mandatory fields (e.g., Date of Birth, Sum Assured) blank.->4. Click "Proceed". | System highlights each missing mandatory field with an inline error message (e.g., "Date of Birth is required"). Purchase cannot continue until fields are completed. |

ITC_IP_004 | OTP timeout handling during policy purchase | User is logged in; user has a verified mobile number; premium within allowed range | 1. Initiate a new policy purchase and reach the OTP screen.->2. Wait for OTP expiration period (e.g., 2 minutes) without entering OTP.->3. Attempt to submit an empty OTP field.->4. Click "Resend OTP". | System shows "OTP expired" message, disables the submit button, and sends a new OTP after "Resend OTP" is clicked. |

ITC_IP_005 | Payment gateway failure during policy purchase | User is logged in; user has a verified mobile number; premium within allowed range | 1. Start a new policy purchase and complete all required steps up to payment authorization.->2. Simulate payment gateway error (e.g., network timeout or declined transaction).->3. Click "Confirm Purchase". | System displays an error message "Payment could not be processed. Please try again or use a different payment method." The policy is not created and no charge is applied. |

Generate test case for account opening

Test Case ID | Title | Preconditions | Steps | Expected Result |

-----|-----|-----|-----|-----|

ITC_AB_001 | Successful account opening with valid details | User is on the "Open New Account" page; all required services are up | 1. Enter a unique customer name.->2. Enter a valid date of birth (age >= 18).->3. Provide a correctly formatted email address.->4. Input a valid mobile number.->5. Select account type (e.g., Savings).->6. Upload required identity documents (e.g., PAN, Aadhaar).->7. Click "Submit". | Account is created, a confirmation message with account number is displayed, and a welcome email/SMS is sent. |

ITC_AB_002 | Account opening fails due to missing mandatory fields | User is on the "Open New Account" page | 1. Leave the "Customer Name" field blank.->2. Fill all other fields with valid data.->3. Click "Submit". | System displays a validation error "Customer Name is required" and prevents submission. |

ITC_AB_003 | Account opening fails with invalid email format | User is on the "Open New Account" page | 1. Enter a valid customer name.->2. Enter a valid date of birth.->3. Input an email address without "@" (e.g., usrexample.com).->4. Fill remaining fields with valid data.->5. Click "Submit". | System shows an inline validation message "Enter a valid email address" and blocks the request. |

ITC_AB_004 | Account opening rejected for age below minimum limit (boundary condition) | User is on the "Open New Account" page | 1. Enter a valid customer name.->2. Enter a date of birth that makes the applicant 17 years old.->3. Fill all other fields with valid data.->4. Click "Submit". | System returns an error "Applicant must be at least 18 years old" and does not create the account. |

ITC_AB_005 | Account opening fails when backend service is unavailable | User is on the "Open New Account" page; backend account creation service is down (simulated) | 1. Enter all required fields with valid data.->2. Click "Submit". | System displays a generic error message "Unable to process request at this time. Please try again later," and logs the failure for investigation. |

Ask me to generate test cases... e.g. 'Generate test cases for fund transfer'

Send

Response received.

The user can toggle between light and dark mode to change the UI color theme

CaseForge AI

Quick Generate

Click a domain to reveal its frequently used workflows.

Netbanking

Financial Services

Insurance

New policy purchase

Claim filing and tracking

Premium payment

Policy renewal

Nominee addition / update

Policy document download

in linked payment method | 1. Navigate to "Buy New Policy".->2. Select "Health Insurance".->3. Choose a plan and enter required personal details.->4. Enter a premium amount within the allowed range (e.g., ₹5,000-₹50,000).->5. Confirm policy terms and click "Proceed".->6. Receive OTP on registered mobile, enter OTP.->7. Authorize payment.->8. Click "Confirm Purchase". | Policy is created, payment is captured, and a confirmation screen with policy number and receipt is displayed. |

ITC_IP_002 | Purchase fails when premium exceeds maximum allowed limit | User is logged in; user has a verified mobile number | 1. Navigate to "Buy New Policy".->2. Select "Motor Insurance".->3. Enter vehicle details.->4. Input a premium amount above the maximum allowed (e.g., ₹2,00,000 when limit is ₹1,50,000).->5. Click "Proceed". | System displays a validation error: "Premium exceeds maximum allowed limit" and prevents further progression. |

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Generate test case for account opening

Test Case ID | Title | Preconditions | Steps | Expected Result |

-----|-----|-----|-----|-----|

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ITC_AB_005 | Account opening fails when backend service is unavailable | User is on the "Open New Account" page; backend account creation service is down (simulated) | 1. Enter all required fields with valid data.->2. Click "Submit". | System displays a generic error message "Unable to process request at this time. Please try again later," and logs the failure for investigation. |

Ask me to generate test cases... e.g. 'Generate test cases for fund transfer'

Send

Response received.